

The calculator was designed using separation of concerns, where Calculator handles arithmetic logic and CalculatorUI manages the graphical interface. A state-machine approach was used to simplify input handling, operator chaining, and error management.

Main functionalities include addition, subtraction, multiplication, division, modulo, decimal input, negation, delete, clear, chained operations, and error handling such as division by zero.

The implementation uses Java Swing for the interface. Calculator.java performs arithmetic operations, while CalculatorUI.java creates the display, buttons, and event handlers. Main.java launches the application using SwingUtilities.invokeLater() for thread-safe GUI execution.